

Task 01: Contention Window Size

Observation Table

Metric	cwMin = 3	cwMin = 15	cwMin = 63
Packet animations (sparse/dense?)	Dense	Moderately dense	Sparse
Noticeable pauses / backoff delays?	Very short pauses but frequent; rapid re-attempts after collisions	Occasional brief pauses; backoff delays are noticeable but balanced	Long, regular pauses between transmissions; backoff delay clearly visible
Console: retransmission messages?	Frequent retransmission warnings	Occasional retransmissions	Rare retransmissions;
Qualitative throughput	Low	Medium–High	Medium

Analysis Questions

a. Why does smaller cwMin increase collision probability?

In CSMA/CA, before transmitting, each station draws a random backoff value uniformly from the range $[0, \text{cwMin}]$. With $\text{cwMin} = 3$, the entire pool of possible backoff slots is only $\{0, 1, 2, 3\}$ — four values shared among four competing hosts. The probability that at least two hosts independently select the same slot, and thus transmit simultaneously causing a collision, is very high.

b. How many distinct backoff values with $cwMin = 3$?

0, 1, 2, 3

c. Which $cwMin$ gave the best performance?

$cwMin = 15$

d. Why isn't a larger $cwMin$ always better in a dense network?

A larger $cwMin$ means every host waits longer on average before each transmission attempt, even when few or no competing hosts are present. This introduces **unnecessary idle slots**.

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Task 02: RTS/CTS Effect

Analysis Questions

a. Full RTS/CTS exchange sequence and frame count

Four frames are exchanged before a data frame is fully delivered:

Time →



Frame	Direction	Purpose
RTS (Request to Send)	Sender → AP	Reserve channel; carries NAV duration
CTS (Clear to Send)	AP → All	Silences all other stations for NAV duration
DATA	Sender → AP	Actual payload
ACK	AP → Sender	Confirms successful reception

Total: 4 frames for one successful data delivery (compared to 2 for the basic access method: DATA + ACK).

b. Observed difference in Q_{tenv} between RTS Off and RTS On

With RTS_Off: packet animations show frequent simultaneous transmissions from multiple hosts, with visible collision events and retransmission attempts. The animation is dense and somewhat chaotic. (visible in SS)

With RTS_On (1mW, rtsThreshold=100B): each successful data delivery is preceded by the visible RTS/CTS mini-exchange, which briefly silences other hosts. The net effect is a reduction in mid-data collisions.

c. When is RTS/CTS overhead worth paying? When is it not?

Worth paying

Hidden terminal scenario — stations cannot hear each other but can hear the AP; CTS prevents them from transmitting over an ongoing transfer

Large packets — cost of one collision is much higher than 3 short control frames

Dense, high-contention networks — many competing stations

Not worth it

Small packets — when frame size $\approx$ RTS size, the overhead (3 extra frames) exceeds any collision savings

Low-contention network — if only 1–2 stations, collisions are rare; RTS/CTS just adds latency

Delay-sensitive or real time traffic — VoIP, real-time gaming; the extra round-trip of RTS/CTS adds measurable latency

Task 03: 802.11b vs 802.11g Timing

Calculation Table

Assumed: SIFS = 10 μ s (802.11b), 16 μ s (802.11g OFDM); ACK $\approx$ 14 bytes.

Metric	802.11b	802.11g
Frame duration (1000B at bitrate)	$8000 \text{ bits} \div 11 \text{ Mbps} = \mathbf{727.3 \mu s}$	$8000 \text{ bits} \div 54 \text{ Mbps} = \mathbf{148.1 \mu s}$
DIFS = SIFS + 2 $\times$ SlotTime	$10 + 2 \times 20 = \mathbf{50 \mu s}$	$16 + 2 \times 9 = \mathbf{34 \mu s}$
Avg backoff = (cwMin/2) $\times$ SlotTime	$(31/2) \times 20 = \mathbf{310 \mu s}$	$(15/2) \times 9 = \mathbf{67.5 \mu s}$

Total channel time per frame (approx.)	$50 + 310 + 727.3 + 10 + 10.2 \approx \mathbf{1107 \mu s}$	$34 + 67.5 + 148.1 + 16 + 10 \approx \mathbf{276 \mu s}$
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Max theoretical frames/s (1 / total)	$1 \div 0.001107 \approx \mathbf{904 \text{ frames/s}}$	$1 \div 0.000276 \approx \mathbf{3623 \text{ frames/s}}$
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(ACK $\approx 112 \text{ bits} \div 11 \text{ Mbps} \approx 10.2 \mu s$ for b; $\approx 10 \mu s$ for g)

Analysis Questions

a. How many times faster is a single 802.11g frame delivered vs 802.11b?

802.11g delivers the same 1000B packet approximately **4 times faster** per frame, primarily due to the higher bitrate (4.9× faster frame transmission) compounded by the shorter slot time (9 μs vs 20 μs) reducing DIFS and backoff overhead.

b. Visual difference in Qtenv animation rate?

No visual difference clearly shown in Qtenv animation rate. Maybe slowing the animation may yield good results as the difference is not significant.

c. Why does 802.11b use larger cwMin = 31 despite being slower?

Precisely *because* it is slower. At 11 Mbps, each data frame takes $\sim 727 \mu s$ to transmit — roughly 5× longer than at 54 Mbps. A collision at 11 Mbps wastes far more channel time per event than a collision at 54 Mbps. To keep the collision rate low enough that the already-slow physical layer isn't constantly retransmitting, the standard chose a larger cwMin=31 to spread backoff choices across more slots and reduce collision probability.

d. Which standard for stationary, close-proximity devices?

802.11g parameters are preferable. Key reasons:

- Devices are close - signal quality is high
- Stationary devices - stable channel
- Close proximity - no hidden terminal problem
- Smaller cwMin=15 - lower backoff delay

Task 04: Binary Exponential Backoff

Analysis Questions

a. CW doubling sequence in BEB_Aggressive (cwMin=15, cwMax=63, retryLimit=3)

BEB doubles the contention window after each failed attempt (no ACK received):

Attempt #	Event	CW size
1st transmission	Initial attempt	cwMin = 15
Collision 1 → Attempt 2	CW doubles: $2 \times (15 + 1) - 1$	31
Collision 2 → Attempt 3	CW doubles: $2 \times (31 + 1) - 1$	63 (= cwMax, capped)
Collision 3	retryLimit=3 reached → frame dropped	—

After 3 collisions, the CW reached 63 (the cap) and the frame is discarded by the MAC layer.

b. What happens after 7 failed attempts in BEB_Patient?

After 7 failed transmission attempts, the MAC layer permanently discards the frame and increments the station's drop counter. Since the application is using UDP (UdpBasicApp), there is no transport-layer retransmission mechanism — UDP provides no delivery guarantees. The consequence for the application is silent packet loss: the sender's app continues generating new packets at 0.02s intervals, unaware the previous packet was never delivered. The receiver's echo app never processes that packet, and no notification is sent back to the sender.

c. Qtenv comparison: continuous flow vs. pauses

BEB_Aggressive shows more visible pauses and chaotic bursts. Because cwMax=63 is small, hosts that collide never back off very far; they retry quickly, collide again, and after only 3 attempts the frame is dropped.

d. Risk and benefit of small cwMax in a congested network

Detail

Risk In a heavily loaded network with many competing hosts, a small cwMax means the contention window stops growing after only a few doublings. Hosts continue colliding at roughly the same rate indefinitely because backoff values never spread far enough to reliably separate them.

Benefit When the network is moderately loaded, a small cwMax prevents hosts from backing off excessively after an isolated collision. Recovery is fast — hosts retry sooner, keeping latency low. For real-time applications (VoIP, gaming), a bounded backoff ceiling ensures a maximum bound on retry delay, which is preferable to unbounded exponential growth

e. Proposed optimal parameters for 6-host scenario

Parameter	Proposed Value	Justification
cwMin	15	Standard 802.11g value; 16 backoff slots are sufficient to meaningfully separate 6 hosts on first attempt
cwMax	255	Allows 4 doublings (15→31→63→127→255) before capping; wide enough to reduce sustained collision probability without backing off for hundreds of milliseconds

retryLimit	5	Gives enough attempts for a frame to succeed even under moderate congestion, while not holding channel resources indefinitely for a frame that will likely fail regardless
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Justification: cwMin=15 keeps initial latency low. cwMax=255 (rather than 63) allows BEB to do its job — spreading retrying hosts across a wider window after each collision.